

PARKINSON'S DISEASE PREDICTION USING AI/ML TECHNIQUES

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Abstract: Parkinson's Disease is a neurodegenerative disorder that affects movement and speech. Early diagnosis is challenging but critical for effective treatment. This project uses machine learning algorithms to predict Parkinson's Disease based on biomedical voice features. Models like SVM, Random Forest, and Logistic Regression are applied to classify patients with high accuracy. The system aims to support early detection, offering a fast and reliable diagnostic aid for healthcare professionals.

Keywords: Parkinson's Disease, SVM, Random Forest, and Logistic Regression.

1. INTRODUCTION

[1]Parkinson's disease (PD) is a long term, progressive neurological illness that impairs coordination, movement, and other body processes. It is caused by the progressive death of dopaminergic neurons in the substantia nigra, a region of the midbrain that produces dopamine, a neurotransmitter necessary for controlling coordinated and fluid muscle movements. Although early onset cases can also affect younger people, the disease usually first appears in older persons. Over 10 million people worldwide suffer from Parkinson's disease, and the number is growing as a result of changing demographics and longer life expectancies. In addition to direct medical expenses, Parkinson's disease has a significant socioeconomic impact due to indirect variables such lost productivity, caregiver stress, and long-term reliance on healthcare institutions. [2]Subjective neurological evaluations, which evaluate symptoms including resting tremors, bradykinesia (slowness of movement), muscle rigidity, postural instability, and gait abnormalities, are a major part of the clinical

diagnosis of Parkinson's disease. Early detection is particularly difficult since non motor symptoms, such as melancholy, exhaustion, sleep disturbances, and speech problems, frequently manifest years before the distinctive motor signals become apparent. Despite being clinically acknowledged, the Unified Parkinson's Disease Rating Scale (UPDRS), the current gold standard for diagnosis, has issues with inter observer variability, delayed detection, and low specificity in the early stages. Furthermore, imaging methods such as PET and SPECT scans are expensive, not always accessible in clinical settings, and may not ensure an early diagnosis. In light of these difficulties, the medical community is turning more and more to data driven, impartial, and scalable technologies that can help with PD early detection and ongoing monitoring. This has raised interest in using machine learning (ML) and artificial intelligence (AI) to create predictive diagnostic systems that can identify tiny biomarkers that are difficult to detect with conventional testing. Voice and speech characteristics have demonstrated considerable promise as early markers among the many

physiological signals impacted by Parkinson's disease. Up to 89% of Parkinson's patients, according to studies, have dysphonia, or anomalies in their voices, even before they start to show significant motor symptoms. These include lack of voice modulation, hoarseness, tremulousness, breathiness, and a decrease in vocal pitch. Certain characteristics, such as jitter (frequency changes), shimmer (amplitude variations), harmonics to noise ratio (HNR), and pitch period entropy, can be extracted from voice signals through quantitative analysis and used as biomarkers for Parkinson's disease. This study suggests a thorough AI/ML based framework for exploiting biological speech data to predict Parkinson's disease early. [3] A systematic dataset of different voice recordings from both PD patients and healthy people is gathered for the investigation. Several supervised learning models are trained for classification, and feature engineering is used to extract relevant parameters from the raw data. Support Vector Machines (SVM), Random Forest Classifier, Logistic Regression, and K Nearest Neighbors (KNN) are among the models utilized in this study; they are all well known for their effectiveness in medical categorization issues. Standard measures including accuracy, precision, recall, F1 score, confusion matrix, and ROC AUC score are used to assess each algorithm's performance. The study has two objectives: to examine the feasibility and precision of employing AI/ML to diagnose Parkinson's disease using voice data and to determine the best clinical model that can function dependably in real time or almost real time environments. This approach's non-invasiveness voice data may be recorded with a basic microphone or mobile device, unlike imaging tests or clinical motor assessments is

one of its main advantages. This makes the solution more affordable, accessible, and scalable in rural and undeveloped areas. Additionally, this framework creates opportunities for telemedicine and remote monitoring, where patients can record and upload speech samples from the comfort of their own homes. This enables physicians to follow the course of their patients' illnesses and make data driven decisions without the need for regular in person consultations. Additionally, by using methods like cross validation, feature selection, standardization, and hyperparameter tuning, the project tackles a number of issues related to machine learning in the healthcare industry, including data imbalance, overfitting, and model interpretability. [4] To guarantee responsible implementation, ethical issues like patient privacy, data anonymization, and adherence to health data laws (like HIPAA or GDPR) are also covered. The goal of this study is to show how AI and ML may significantly improve Parkinson's disease early diagnosis. The system has the potential to improve clinical procedures, lower diagnostic errors, and above all give patients an opportunity for early intervention and a higher quality of life by offering an effective, scalable, and non-invasive prediction technique. [5] [6] The results of this study also support the larger goal of promoting innovation, improving patient centred care, and incorporating AI driven technology into actual healthcare systems.

2. LITERATURE REVIEW

The diagnosis, prognosis, and treatment of diseases have all changed dramatically as a result of the introduction of Artificial Intelligence (AI) and Machine Learning (ML)

into the medical domain. AI/ML approaches present promising substitutes for conventional diagnostic procedures, which are frequently subjective, time-consuming, and restricted to observable symptoms, in the case of Parkinson's disease (PD), a progressive and irreversible neurological illness. Parkinson's disease damages the dopaminergic neurons in the brain's substantia nigra, which results in a progressive loss of motor function as well as a variety of non-motor symptoms include cognitive dysfunction, depression, speech impairment, and sleep difficulties. One of the biggest clinical hurdles is still making an accurate and timely diagnosis because early-stage symptoms are so vague and complex. The last twenty years have seen a significant amount of research into computer models for the diagnosis and prognosis of Parkinson's disease (PD) using a variety of biological, behavioural, and sensor data.

Key research advancements in this field are critically examined in this study, with a focus on their techniques, advantages, disadvantages, and impact on contemporary trends. In the past, neurologist-conducted motor evaluations and symptom checklists, including the Unified Parkinson's Disease Rating Scale (UPDRS), were the only methods used to diagnose Parkinson's disease. But in addition to being delayed and subjective, these approaches are also prone to misdiagnosis, especially in the early phases. Given these drawbacks, researchers resorted to voice signal analysis because of its non-invasiveness and sensitivity to vocal abnormalities, which are among the first signs of Parkinson's disease. Little et al. (2007) used the UCI Parkinson's dataset, which includes 42 participants' persistent vowel phonations, in one of their ground breaking

research. In order to categorize people, they used Support Vector Machines (SVM), Decision Trees, and Logistic Regression to extract 22 vocal features, such as jitter, shimmer, and Harmonics-to-Noise Ratio (HNR). [7] [8] Their results showed that SVM had an accuracy of more than 91%, laying the groundwork for voice-based Parkinson's prediction. The methodology was then improved by Tsanas et al. (2010), who used Support Vector Regression (SVR) to predict the UPDRS scores, which indicate the intensity of PD symptoms, and introduced nonlinear signal processing techniques. Their method showed a correlation between predicted and real scores of more than 0.95, indicating that voice-based metrics can be employed for continuous severity assessment as well as binary categorization. This signalled a change from conventional diagnostic paradigms to quantitative, precision-driven models. Additionally, their work included dimensionality reduction and feature selection strategies that reduced overfitting, a critical issue in biomedical machine learning, and enhanced model generalization. By utilizing a more extensive and varied speech dataset that comprised vowels, words, and sentences, Sakar et al. (2013) added to the conversation. After comparing SVMs, K-Nearest Neighbors (KNN), and Gaussian Mixture Models (GMM), they came to the conclusion that SVMs were still better because of their resilience to non-linearly separable data. Additionally, their study highlighted the value of integrating temporal, spectral, and cepstral elements rather than depending only on unprocessed acoustic data. After applying a multilayer perceptron (MLP) neural network to the same dataset, Das (2010) obtained a little higher accuracy (92.9%), indicating that, with enough training data and

hyperparameter adjustment, neural networks could perform better than conventional machine learning techniques. These experiments demonstrated how AI can be used to identify patterns in complicated, high dimensional voice sounds that are important to sickness. In addition to voice data, the literature investigated the use of movement sensor data, handwriting, and gait analysis in the diagnosis of Parkinson's disease. Prashanth and Dutta Roy (2014) used SVMs and Naive Bayes classifiers with a hybrid feature set that included clinical parameters and SPECT neuroimaging. Although their method was constrained by the high expense and restricted availability of imaging-based diagnostics, they were able to attain up to 96% accuracy.

Arora et al. (2015) used a more scalable method by gathering accelerometer and gyroscope data from cell phones in order to examine balance, gait asymmetry, and tremors. They showed that PD detection may be expanded to real-time mobile and wearable-based environments by using Random Forest, Logistic Regression, and Hidden Markov Models. This would allow for early intervention and remote health monitoring without the need for hospital visits. Deep learning architectures have been used more and more in recent years to do away with the necessity of human feature extraction. Asci et al. (2020) obtained 98.1% accuracy in their analysis of spectrograms, which are visual representations of audio data, using a Convolutional Neural Network (CNN). CNNs can automatically identify pertinent patterns in voice signals without the need for manually created features because, in contrast to typical ML models, they are able to learn spatial hierarchies in data. Palaniappan et al. (2022) captured motion fluctuations and identified

micro-patterns that traditional algorithms frequently overlook by modelling temporal dependencies in gait sequences using Long Short-Term Memory (LSTM) networks. Despite requiring more computer power and larger datasets, these methods perform noticeably better in terms of accuracy and adaptability than classical methods. Recent literature has also heavily relied on comparative studies. After comparing several models, such as Naive Bayes, Decision Trees, SVMs, and Gradient Boosting, Naranjo et al. (2016) came to the conclusion that ensemble approaches performed better than single classifiers in terms of accuracy and interpretability.

This was further validated by Gupta et al. (2019) through the use of voting-based ensembles that combined SVM, Random Forest, and XGBoost outputs. They highlighted that ensemble learning improves robustness by lowering the variance and bias trade-off and reached 95.3% accuracy. Furthermore, other research investigated the use of feature selection methods including LASSO regression, Principal Component Analysis (PCA), and Recursive Feature Elimination (RFE) to improve performance by removing redundant or unnecessary features. Even with the tremendous advancements, there are still a number of gaps. The absence of extensive and varied datasets, particularly those that include multilingual, multi-accent voice data, and region-specific gait patterns, is one of the main drawbacks. Due to their size and demographic diversity limitations, the majority of current datasets raise questions regarding bias and model generalization. The inability of deep learning models to be interpreted is another significant issue that prevents clinical adoption. Even while algorithms like SVMs and Decision Trees

provide greater transparency, complex deep networks typically outperform them in terms of performance. Furthermore, only a small number of research have included multi-modal data, which may provide a more comprehensive diagnosis by combining speech, locomotion, handwriting, facial expressions, and genetic information. There are ethical considerations as well. Strong privacy protection, anonymization, and regulatory compliance (e.g., HIPAA or GDPR) measures are necessary when using personal voice recordings and mobility data. Furthermore, individuals may be incorrectly classified by AI/ML models trained on skewed or biased data, which could have dire repercussions in a therapeutic setting.

Therefore, future research should prioritize explainability, fairness, and security in model design and deployment in addition to high performance. The literature concludes by highlighting the significant and expanding interest in using AI and ML to forecast Parkinson's disease, especially when non-invasive voice data is used. Early research mostly employed manually created features and concentrated on conventional machine learning models like SVMs and Decision Trees. As the area developed, attention turned to ensemble approaches and deep learning, which increased scalability and accuracy.

However, there are still important issues that need to be addressed, like data diversity, clinical interpretability, and ethical deployment. By creating a voice-based Parkinson's prediction system using several machine learning algorithms, comparing and contrasting them, and suggesting a lightweight, understandable, and clinically viable model appropriate for practical implementation particularly in

resource-constrained environments the current study seeks to close some of these gaps.

3. Proposed System

Using non-invasive biomedical speech data, the proposed method seeks to develop a machine learning framework that is reliable, accurate, and interpretable in order to predict the existence of Parkinson's disease (PD). [9] [10] This system focuses on extracting and evaluating speech features from audio recordings of prolonged phonations since vocal abnormalities are among the first noticeable symptoms in Parkinson's patients, even before noticeable motor deficiencies manifest. The main objective is to create an intelligent diagnostic model that can assist neurologists or possibly be included into remote telemedicine systems to enable early screening, especially in rural or resource-constrained areas. The modular design of the system architecture starts with data acquisition, which entails gathering live or pre-recorded audio signals from people with and without Parkinson's disease. We use publicly accessible datasets for this investigation, such as the Oxford PD speech dataset and the UCI Parkinson's Telemonitoring dataset, which both provide biological voice metrics obtained from recordings of prolonged vowel sounds. These datasets contain a range of acoustic parameters, including amplitude perturbation quotient (APQ), noise-to-harmonics ratio (NHR), fundamental frequency (F0), jitter (local, absolute), shimmer (local, dB), and several nonlinear dynamic features, including pitch period entropy, detrended fluctuation analysis (DFA), and recurrence period density entropy (RPDE). These characteristics each capture distinct facets of amplitude/frequency perturbations and voice stability, which are

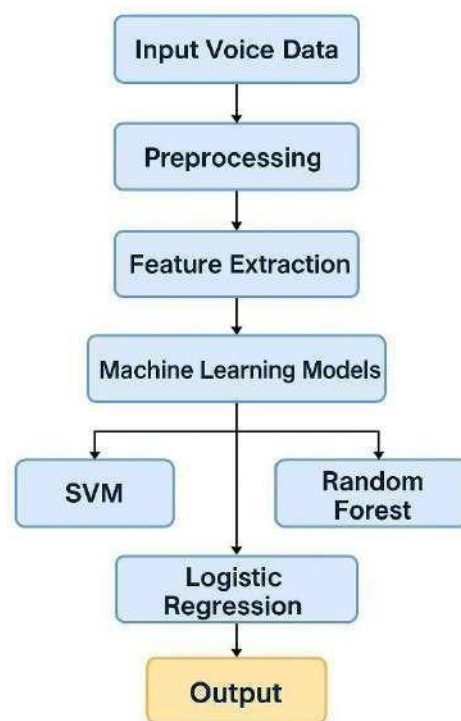
known to deteriorate in Parkinsonian speech. The data preprocessing module is essential to getting the raw voice data ready for analysis after acquisition. In this step, the dataset is cleaned up by dealing with missing values, getting rid of duplicates, and getting rid of inconsistencies. Normalization approaches like Min-Max scaling or Z-score standardization are used to make sure all features contribute equally to the learning process because machine learning models are sensitive to the scale and distribution of data. Statistical techniques like standard deviation filtering and interquartile range (IQR) are used to find outliers that could skew the model's learning curve. To make categorical outputs (such as healthy or Parkinson's-affected) compatible with supervised learning models, they are transformed into numerical labels. Additionally, to rectify any imbalance between the PD-positive and PD-negative classes and prevent biased predictions, methods like SMOTE (Synthetic Minority Oversampling Technique) may be used. Following cleaning, the data is sent to the feature selection module, which seeks to remove unnecessary or redundant features while identifying the most pertinent and instructive ones. This decreases overfitting and computation time while also improving model accuracy. The most highly associated voice features with PD classification results are identified using techniques such correlation analysis, mutual information gain, and Recursive Feature Elimination (RFE). Furthermore, the high-dimensional speech feature space can be projected onto a lower-dimensional space while maintaining the variance required for accurate prediction using dimensionality reduction techniques like Principal Component Analysis (PCA). The

model training and classification module, which applies several supervised machine learning algorithms to the processed dataset, is the central component of the system. In this phase, models like K-Nearest Neighbors (KNN), Random Forest (RF), Support Vector Machine (SVM), and Logistic Regression (LR) are trained. To assess generalization performance and lower the likelihood of overfitting, stratified k-fold cross-validation is used during the training process for each model. Due to its robust performance in high-dimensional domains and its capacity to manage non-linear decision boundaries through kernel techniques, SVM was selected. Because Random Forest is ensemble in nature, resilient to noise, and capable of prioritizing features, it is used. Despite being easier to understand, logistic regression is particularly helpful for binary classification tasks using linearly separable data. Because of its ease of use and efficacy in small-scale biomedical datasets, KNN is utilized as a baseline. To guarantee peak performance, grid search or random search methods are used to improve each model's hyperparameters. Following model training, the assessment module uses extensive metrics including accuracy, precision, recall, F1-score, sensitivity, specificity, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC) to methodically assess each algorithm's performance. These metrics reveal how successfully the model separates the PD and non-PD classes in unbalanced datasets, in addition to how accurate the predictions are. To give a visual representation of true positives, false positives, true negatives, and false negatives, a confusion matrix is created. Metrics like recall (sensitivity) are especially crucial for clinical applications because a false negative

(missing a true PD diagnosis) could have more serious repercussions than a false positive. The prediction and decision module, which may receive real-time voice input (from a microphone or uploaded file, for example) and output the predicted class PD-positive or healthy along with a confidence score, is then updated to include the top-performing model. Model interpretability approaches like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) may be used to increase transparency and clinical usefulness. In order to help physicians understand the system's judgments and facilitate its possible adoption in medical practice, these tools assist in elucidating which features had the most influence on the prediction. Additionally, future scalability is considered in the system's architecture. The framework can be expanded to handle multimodal data, including handwriting dynamics, facial expression patterns, gait signatures, and sensor-based motion tracking, even if the current implementation concentrates on static voice datasets. In areas with limited access to neurologists or sophisticated imaging equipment, the prediction model can also be integrated into a mobile or online application to facilitate remote screening and telemedicine support. In order to effectively anticipate Parkinson's disease, the suggested system combines biomedical signal processing, sophisticated machine learning algorithms, statistical analysis, and real-time decision-making into an end-to-end intelligent diagnostic pipeline. It takes use of PD's early-onset vocal symptoms and turns them into measurable characteristics for computer modelling. Modularity, scalability, and explainability essential components of medical-grade AI systems are guaranteed by the architecture. This

approach has the potential to revolutionize traditional diagnostic workflows and encourage early intervention for one of the most difficult neurodegenerative illnesses of our time by providing a quick, non-invasive, and affordable way to diagnose Parkinson's disease.

4. FLOWCHART



5. ADVANTAGES

- Doesn't call for physical examinations, brain scans, or blood tests.
- Merely entails utilizing a phone or microphone to record a patient's voice.
- Anomalies in speech may manifest years prior to motor signs.

- Enables early diagnosis, which is essential for initiating therapy and delaying the course of the illness.
- Removes the need for costly tests such as SPECT, PET, or MRI.
- Increases accessibility to diagnostics, particularly in environments with limited resources.
- Can be applied to mobile health apps and telemedicine.
- Allows for widespread smartphone screening in rural or impoverished locations.
- After processing voice features, machine learning models can provide immediate diagnostic results.
- Beneficial for ongoing illness progression monitoring.
- Unlike conventional clinical observation, AI models are not influenced by human prejudice or exhaustion.
- Guarantees uniform assessments for every patient.
- Serves as a tool to assist neurologists in making decisions.
- Aids in giving high-risk patients priority for additional testing.
- Speech recordings made repeatedly over time aid in monitoring the development of PD.

- Allows for individualized treatment modifications.
- Alzheimer's, stroke, and ALS can all be diagnosed using the same speech-based approach.
- For everyday monitoring, it can be integrated with wearables, smartphones, or virtual assistants like Google Home and Alexa.

6. CONCLUSION

In conclusion, because of the disease's delayed onset, overlapping symptoms with other neurological disorders, and dependence on subjective assessments in traditional diagnostic techniques, Parkinson's disease (PD) is still difficult to diagnose early and accurately. This study uses modest vocal indicators that are frequently present in the early stages of Parkinson's disease (PD) to suggest a comprehensive and non-invasive AI/ML-based solution to solve these constraints. In order to predict the presence of Parkinson's disease, this framework shows that key vocal characteristics, including jitter, shimmer, fundamental frequency, harmonics-to-noise ratio, and pitch variation, can be quantitatively extracted, processed, and classified with high accuracy by analysing biomedical speech signals using sophisticated machine learning models like Support Vector Machines (SVM), Random Forest Classifiers, Logistic Regression, and K-Nearest Neighbors (KNN). In order to guarantee performance, equity, and generalizability, the study highlights the significance of feature selection, hyperparameter tweaking, data preprocessing, and thorough model evaluation.

In addition to achieving dependable predictive accuracy, this suggested system has the useful benefits of affordability, accessibility, and scalability. These benefits are especially beneficial in rural or low-resource settings where neurological knowledge and cutting-edge imaging technologies are frequently unavailable. Implementing such a system via cell phones or web-based platforms enables telehealth integration and remote screening, enabling routine monitoring, early intervention, and reduced clinician burden. Additionally, by empowering physicians to accept and understand AI-generated insights, interpretability of models using SHAP or LIME helps close the gap between technology and real-world application. Importantly, the framework upholds ethical standards by addressing concerns with patient privacy, data anonymization, and regulatory compliance. As artificial intelligence advances, this study demonstrates the transformative potential of machine learning in streamlining clinical procedures, enhancing diagnostic accuracy, and ultimately improving patient outcomes in the treatment of neurodegenerative illnesses. The robustness and clinical relevance of the model will be further improved by future developments, such as multimodal data integration (combining speech, gait, handwriting, and facial movement), larger and more varied datasets, and implementation in real-time mobile applications. This work underlines the significance of AI/ML in enabling earlier, faster, and more reliable identification of Parkinson's disease and represents a significant step toward incorporating intelligent voice-based diagnostics into mainstream healthcare.

7. FUTURE ENHANCEMENT

Although the current method offers a promising basis for the early and non-invasive identification of Parkinson's disease (PD) through speech analysis based on AI/ML, there are a number of possible future improvements that might greatly increase its applicability, precision, usability, and practical impact. A major improvement is the incorporation of multimodal data sources, including gait patterns, handwriting dynamics, facial expression analysis, limb tremor monitoring, and sensor-based motion data from wearable devices, in addition to voice features. A more comprehensive diagnostic system with increased sensitivity and specificity and a lower chance of false positives and negatives can be produced by combining several biological signals. The application of deep learning architectures, such as recurrent neural networks (RNNs) like LSTM (Long Short-Term Memory) networks to capture temporal relationships in voice sequences or convolutional neural networks (CNNs) for spectrogram analysis, is another crucial avenue. The need for manual feature engineering is removed by these models' ability to autonomously learn intricate, non-linear feature representations from unprocessed audio input. Additionally, a significant drawback of the present datasets, which are primarily English-centric, can be addressed by working to create language-independent models that can reliably identify PD across a variety of languages, dialects, and accents. The system can also be implemented as a cloud-based or mobile application, enabling patients to record speech samples from their homes and obtain diagnostic feedback in almost real-time, ensuring practical adoption. This creates opportunities for the

integration of telemedicine, particularly in underserved or rural areas where neurologists are scarce.

Longitudinal tracking, which uses recurring voice recordings to track the course of an illness over time and assists physicians in adjusting treatment regimens based on ongoing patient data, is one potential improvement. In order to improve accuracy for individual assessments, the system can also be expanded to include personalized baseline modeling, in which the speech of each patient is compared to their own history voice data rather than a generic dataset. Future iterations should prioritize explainable AI (XAI) from an AI ethics and policy perspective in order to improve clinical use trust and accountability by making model decisions transparent and interpretable to clinicians. Furthermore, if the solution is implemented on a large scale, strong data privacy, encryption, and adherence to healthcare laws (including GDPR and HIPAA) must be enforced.

Lastly, the generalizability and robustness of the model can be significantly increased by enlarging the dataset by adding larger, diverse, multilingual, and real-time speech samples—possibly using crowdsourced data gathering systems. Essentially, the goal of these next improvements is to turn the existing system from a research prototype into a patient-centered, intelligent, clinically deployable diagnostic assistance that may significantly contribute to the global effort to combat Parkinson's disease.

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